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Modeling of reverse water–gas shift reaction in a membrane integrated microreactor

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ABSTRACT

CO_2 hydrogenation to syngas *via* reverse water–gas shift (RWGS) is modeled in a membrane decorated microchannel reactor. Coated $CuO/ZnO/Al_2O_3$ catalyst exists in the reaction channel which is physically separated from the neighboring permeate channel by a layer of supported sodalite (SOD) membrane selective to H_2O and H_2 transport. Pure H_2 is used as the sweep gas. Two–dimensional, steady state, isothermal operation of the reactor is quantified by the momentum and mass conservations in the entire flow domain, membrane material transfer and catalytic reaction. Upon its benchmarking with the literature–based experimental data, the reactor model is used to study the effects of inlet velocities and partitioning of the reactive mixture and sweep streams, reactor pressure, and molar inlet H_2:CO_2 ratio. At 523 K, 15 bar and H_2:CO_2 = 3, the per cent CO_2 conversion exceeds the thermodynamic limit of 16.8% and can be maximized to 52% when the inlet velocity of the $H_2 + CO_2$ mixture is minimized. Decreased dosing of the reactive mixture, however, limits the molar amount of CO_2 converted and reduces reactor capacity. Both metrics are positively correlated with reactor pressure and the inlet velocity of the sweep gas but remain insensitive to flow direction. Syngas composition (molar exit $(H_2 - CO_2)/(CO + CO_2)$ ratio) can be varied between 1.7 and 9.8. The proposed reactor can deliver ~2.5 times higher volume and catalyst mass–based productivity and ~33% higher CO_2 conversion than an equivalently operated packed–bed membrane reactor.

1. Introduction

Synthesis gas (syngas), a mixture of carbon oxides (CO and CO_2) and H_2, is the platform for important chemicals and fuels such as methanol, Fischer–Tropsch (FT) liquids, and dimethyl ether (DME). The current route to syngas involves steam reforming, autothermal reforming, (ATR) or partial oxidation (POX) of natural gas [1]. While these technologies are suited for large scale productions (*e.g.,* $> \sim 10^5$ barrels/day of FT liquids) to reduce the unit cost of making syngas [2], they inherently escalate the volume of CO_2 emitted to the atmosphere along with other undesirable species. Increased awareness on the negative impacts of the anthropogenic emissions of CO_2 promotes incentives for the technologies enabling its capture and conversion. In this regard, CO_2 is considered as a carbon source that can react with H_2 to several value–added products such as methanol, DME, FT liquids (*e.g.,* sustainable aviation fuel, gasoline, diesel), with the type and the purity of the product depending primarily on the catalyst type and reaction conditions. Availability of the required H_2 *via* solar or wind–powered water electrolysis makes the CO_2 hydrogenation route even more promising in terms of reduced carbon footprint [3–5]. Conditions and catalyst configurations of CO_2 hydrogenation can also be tuned to produce syngas *via* the reverse water–gas shift reaction (RWGS):

$$CO_2 + H_2 \rightleftharpoons CO + H_2O,\ \Delta H^o = 42.1\ \text{kJ mol}^{-1} \quad (1)$$

The reaction is endothermic and thermodynamically favored to produce CO as a major product at temperatures in excess of 973 K depending on the relative amounts of molar inlet CO_2 and H_2 [6,7]. However, RWGS reaction competes with the unwanted exothermic Sabatier reaction that is thermodynamically promoted at lower temperatures:

$$CO_2 + 4H_2 \rightleftharpoons CH_4 + 2H_2O,\ \Delta H^o = -165\ \text{kJ mol}^{-1} \quad (2)$$

Catalyst type plays a crucial role in shaping–up the selectivity towards the reactions. Cu, Pt and Pd–based catalysts generally favor RWGS, whereas Ni, Rh and Ru catalysts are more selective to Reaction (2) [6–8]. Apart from a proper selection of the catalyst, process intensification techniques can be used to alter the competition between the

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Nomenclature

C_{FP}	pressure correction factor, –.
E_a	activation energy, J mol^{-1}.
$F_{i,in}$	inlet/initial molar flow rate of species *i*, mol s^{-1}.
$F_{i,out}$	outlet molar flow rate of species *i*, mol s^{-1}.
J_i	molar flux of species *i* through membrane, mol m^{-2} s^{-1}.
K_{eq}	equilibrium constant for the RWGS reaction, –.
k_0	pre–exponential factor, mol kg$_{cat}^{-1}$ s^{-1}.
p	total reactor pressure, Pa.
p^i_{perm}	partial pressure of species *i* in the permeate channel, Pa.
p^i_{rxn}	partial pressure of species *i* in the reaction channel, Pa.
$Perm_i$	membrane permeability of species *i*, mol m^{-2} s^{-1} Pa^{-1}.
R	gas constant (= 8.314 J mol^{-1} K^{-1}).
r_{RWGS}	rate of RWGS reaction, mol kg$_{cat}^{-1}$ s^{-1}.
T	temperature, K.
$v^p_{rxn,c}$	Linear velocity of the $H_2 + CO_2$ mixture into the reaction channel at pressure *p*. The subscript *c* refers to an index referring to different velocity values (*c* = 1, 2, 3, 4 at 5 and 10 bar operations; 1, 2, …, 8 at 15 bar operation) (m s^{-1}).
W	mass of catalyst, kg.
y_i	mole fraction of species *i*, –.

Greek letters

?H^o	standard heat of reaction, kJ mol^{-1}.

Subscripts and superscripts

cat	catalyst
i	species index

two reactions in favor of the RWGS. In the so–called chemical looping (CL) aided RWGS, oxygen transfer material is used in a way that it is first reduced to transform H_2 into H_2O and then oxidized again by taking oxygen from CO_2 to produce CO [9–11]. Since the reduction and oxidation of the oxygen transfer material occur separately, the possibility of methane formation *via* Reaction (2) is eliminated, and the gas separation is simplified significantly. The conventional catalytic RWGS temperatures of ~973 K can be reduced to ~723-823 K by chemical looping [9-11]. These temperatures can be reduced further by *in–situ* removal of water *via* using a steam–selective adsorbent or a membrane, which assists in widening the thermodynamic window and increases the operability of the catalysts at temperatures as low as ~523 K [12–15]. When this is combined with an active catalyst capable of driving CO_2 hydrogenation selectively to CO formation at low temperatures, the RWGS route can be considered as a promising alternative for making syngas. Owing to its cyclic nature and the periodic demand on heat input for adsorbent regeneration, the sorption–enhanced RWGS is conceptually less practical than the membrane–integrated configuration capable of being operated continuously operation without extra energy need to drive the membrane separation.

The use of steam–selective membranes in catalytic reactions that involve steam as an unwanted product is receiving interest in recent years. Experimental and modeling studies on DME synthesis *via* hydrogenation of CO_2 [16] or CO + CO_2 mixtures [17–21] commonly point out substantial improvement in DME production upon decoration of the catalytic reactor with hydrophilic membranes such as Linde Type A (LTA) or hydroxy sodalite (H–SOD). Significant improvements in membrane assisted CO_2–to–methanol and FT processes are clearly presented in the reviews of van Kampen et al. [12] and Li et al. [13]. Unlike these processes, studies on membrane–assisted RWGS are limited. Dzuryk and Rezaei [22] modeled the RWGS reaction in a packed–bed membrane reactor comprising a $CuO/ZnO/Al_2O_3$ catalyst and a steam–selective membrane. The authors studied the effects of molar inlet H_2/CO_2 ratio, H_2O/H_2 permselectivity, gas hourly space velocity, and relative values of the pressures and flow rates of the sweep (N_2) and reactive mixture in the co– and counter–current modes. They reported that CO_2 conversion could be increased from ~15% to ~60% upon membrane integration at H_2:CO_2 = 3 and at a temperature of 523 K, which was significantly below that of the (*i.e.,* > ~973 K) conventional catalytic RWGS. These findings were coherent with the thermodynamic predictions that showed the possibility of exceeding the pertinent theoretical barrier upon membrane incorporation [23]. The benefits of *in–situ* membrane separation of H_2O in the context of methanol synthesis *via* RWGS were shown by Samimi et al. [24]. Comparison of the membraneless and membrane–integrated scenarios revealed ~4% improvement in methanol production rate in the latter.

The limited number of studies on membrane integrated RGWS operation commonly consider two concentric tubes with the particulate catalyst being packed into the inner tube whose wall functions as a membrane. In this design, surface area of the membrane per reactor volume can be increased by reducing the diameter of the inner tube to fit more of them into the outer tube and/or by increasing the tube length. However, these actions elevate frictional pressure loss in the inner tube [25]. Further volumetric intensification without increasing pressure drop is possible by structuring the flow paths to micro/milli–scales [26]. Decoration of the layers of membrane between the neighboring permeate and catalyst washcoated reaction channels can increase the surface area per volume of the resulting unit by a factor of ~10^2 [27–29]. Moreover, assuming identical feeding to the permeate and reaction channels, and minimization of the variations in the channel dimensions through high precision micro–machining and reproducible catalyst coating techniques, the unit can be scaled up simply by numbering–up the channels according to the desired capacity [30,31]. Miniaturized dimensions can also promote heat exchange between the neighboring channels allowing the possibility of integrating heating or cooling function with the reaction and/or membrane separation [18,19, 26,32]. These features are demonstrated in the computational modeling of a membrane microchannel reactor for CO_2 hydrogenation to DME [16]. In addition to the capability of delivering isothermal conditions, the reactor offered CO_2 conversion and DME yield up to ~73% and ~35%, respectively, both of which were *ca.* three times higher than those obtained in the membraneless counterpart. In the light of these studies reporting the advantages of volumetric and functional intensification on reactor performance, RWGS reaction is comprehensively modeled in a microstructured catalytic membrane reactor. Pure H_2, one of the readily available reactants, is tested as the sweep gas to enable simultaneous efflux of H_2O from and influx of H_2 to the reaction zone for boosting kinetic and thermodynamic promotion of the RGWS. After validating it with the literature–based experimental data, the reactor model is used to understand the effects of inlet velocities and partitioning of the reactive mixture and sweep streams, reactor pressure and molar inlet H_2:CO_2 ratio on reactor performance. The opposing relation between, per cent CO_2 conversion and the amount of CO_2 consumed, the performance metrics used herein, are demonstrated and the optimum set of operating conditions are determined by considering syngas composition as a constraint. Additionally, volume and catalyst–based productivities of the proposed membrane microchannel reactor are compared with those of a packed–bed membrane reactor under equivalent conditions. To our best knowledge, the findings reported herein provide new and useful insight into efficient syngas production with flexible composition at mild operating conditions.

2. Mathematical modeling

The proposed multichannel membrane reactor (Fig. 1a) comprises repeating unit cells of permeate and reaction channels segregated by a membrane layer (Fig. 1b). The porous $CuO/ZnO/Al_2O_3$ catalyst is in the form of a thin layer that is coated to the inner *x*–*z* plane of the reaction

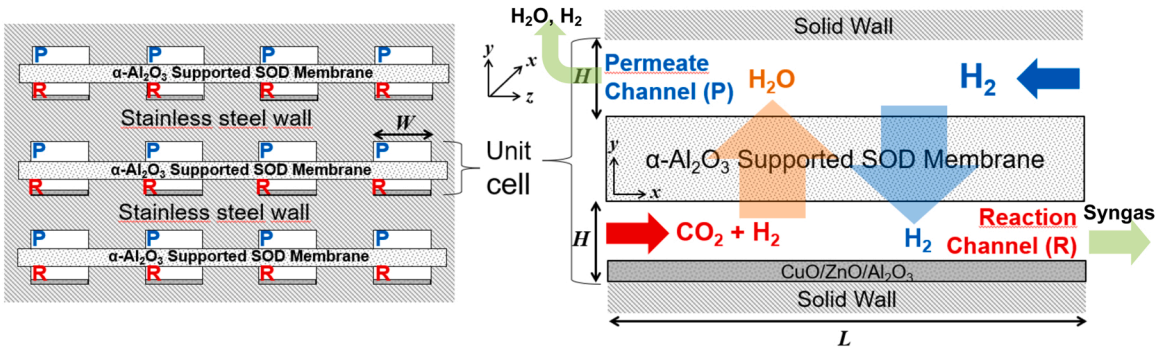


Fig. 1. Schematic illustration of (a) the multichannel reactor unit in two dimensions, (b) the unit cell of the reactor.

channel facing the same plane of the membrane. The thickness of the catalyst layer is taken as 5×10^{-5} m to minimize intra–layer diffusion resistance [29]. Reaction and permeate channels with a rectangular cross–section have the identical dimensions of 3×10^{-4} m (H), 6×10^{-4} m (W) and 1.5×10^{-1} m (L). These dimensions are determined by considering the opposing effects of pressure drop and transport coefficients [33]. Owing to its hydrophilic nature and operability at the current reaction conditions, sodalite (SOD) is selected as the membrane material [34]. In the simulations, SOD is being considered as deposited on 1×10^{-3} m thick α–Al_2O_3 support to form the membrane layer (Fig. 1). The SOD membrane allows simultaneous efflux of H_2O from and influx of H_2 into the reaction channel (Fig. 1b) [34]. Considering its availability in the feed stream, H_2 is also considered as the sweep gas to take the advantage of its transmembrane feeding into the reaction channel for kinetic and thermodynamic promotion of the RWGS reaction.

The default operational settings involve isothermal, co–current flow of the reactive mixture and sweep streams at identical inlet temperature and pressure of 523 K and 15 bar, respectively, at steady state. In all simulations temperature in the entire domain of the unit cell (Fig. 1b) is fixed at 523 K by considering the trade–off between a notable decrease in the rate of the RWGS reaction < 523 K and the increase in the selectivity towards methanation and risk of hydrothermal sintering of the catalyst at higher temperatures [35,36]. Based on the results of $CuO/ZnO/Al_2O_3$ catalyst driven RWGS reaction simulations in the range of 493–523 K [22], default value of the residence time, defined as the ratio of the mass of catalyst to the inlet molar flow rate of CO_2 ($W/F_{CO_2,in}$) is fixed to 40 g_{cat} h mol^{-1}. At this residence time and 523 K, CO_2 conversion is maximized as pointed out by the literature–based experimental data [37]. The default value of the reactor pressure is taken as 15 bar, and the range is set to 5–15 bar to gauge the impact of pressure on *in–situ* membrane separation (Table 1).

Individual effects of H_2:CO_2 and flow direction are investigated by changing the related parameter within its specified range and keeping the others constant at their default values. At 523 K, H_2:$CO_2 = 3$ and the specified pressure (5–15 bar), permeate channel inlet velocity is set to 1, 2, 4 and 6 times the pertinent reaction channel inlet velocity ($v^p_{rxn,c}$) which is changed by adjusting the residence time ($W/F_{CO_2,in}$) accordingly as given in Table 1. In all cases, catalyst layer dimensions and the mass of catalyst coated to one reaction channel (W), calculated by using the physical properties of the $CuO/ZnO/Al_2O_3$ catalyst presented in the Supporting Information (SI), remain constant. Default values of the reaction channel inlet linear velocities at 5, 10 and 15 bar (2.8, 1.4 and 0.94 $\times 10^{-2}$ m s^{-1}, respectively) are based on the residence time of 40 g_{cat} h mol^{-1} and computed by using the ideal gas equation of state. It is assumed that reactive mixture ($H_2 + CO_2$) and sweep (H_2) streams are distributed uniformly to each channel. This assumption is based on the uniformity of channel dimensions and the thickness of the catalyst layer along the channel, both of which can be achieved by the catalyst coating and precise micro–manufacturing techniques [38]. Recent studies pointed out negligible differences in the reactor performance upon setting a pressure gradient between the inlets of reaction and permeate channels [16,17]. In this regard, reactive mixture and sweep streams are always fed at identical pressures. From a practical point of view, this setting prevents operational complexities due to regulation of different inlet pressures. Due to the negligible pressure drop ($< \sim 1\%$) reported for similar fluid flow rates and channel dimensions [39], the fluids in both channels are assumed to be incompressible.

The RWGS reaction is driven only by the coated $CuO/ZnO/Al_2O_3$ catalyst (Fig. 1), and the non–catalytic transformations in the gas–phase are neglected. Considering that the deposition of the surface carbon species, hydrothermal sintering of Cu–nanoparticles and methane formation are observed at temperatures above ~573 K [35,36], Reaction (2) is ruled out and deactivation of the catalyst layer is not considered at the current temperature of 523 K. In the light of these assumptions, RWGS occurs on the $CuO/ZnO/Al_2O_3$ catalyst according to the following rate law [22], whose parameters are provided in SI (Section A):

Table 1
Default values and ranges (presented as boldface) of the operating parameters.

Parameter	Default Value / Setting		Range
T (K)	523		–
$W/F_{CO_2,in}$ (g_{cat} h mol^{-1})	40		**40–320** (40, 80, 160, 320) (at 5 and 10 bar); **5–320** (5, 6.7, 10, 20, 40, 80, 160, 320) (at 15 bar)
p (bar)	15		**5–15** (5, 10, 15)
H_2:CO_2 ratio (–)	3		**2–4** (2, 3, 4)
Flow direction	Co–current		Co/counter–current
Reaction channel inlet velocity at p = 5, 10, 15 bar, $v^p_{rxn,c}$(×10^2 m s^{-1})	5 bar	2.8	**3.5 ×10^{-1}–2.8** ($v^5_{rxn,1}$=3.5 ×10^{-1}; $v^5_{rxn,2}$=7 ×10^{-1}, $v^5_{rxn,3}$=1.4; $v^5_{rxn,4}$=2.8)
	10 bar	1.4	**1.75 ×10^{-1}–1.4** ($v^{10}_{rxn,1}$=1.75 ×10^{-1}, $v^{10}_{rxn,2}$=3.5 ×10^{-1}, $v^{10}_{rxn,3}$= 7 ×10^{-1}, $v^{10}_{rxn,4}$=1.4)
	15 bar	0.94	**1.18 ×10^{-1}–7.5** ($v^{15}_{rxn,1}$=1.18 ×10^{-1}, $v^{15}_{rxn,2}$=2.35 ×10^{-1}, $v^{15}_{rxn,3}$=4.7 ×10^{-1}, $v^{15}_{rxn,4}$=9.4 ×10^{-1}, $v^{15}_{rxn,5}$=1.9, $v^{15}_{rxn,6}$=3.8, $v^{15}_{rxn,7}$=5.6, $v^{15}_{rxn,8}$=7.5)
Permeate channel inlet velocity at p = 5, 10, 15 bar (×10^2 m s^{-1})*	5 bar	2.8	$v^5_{rxn,c}$, $2v^5_{rxn,c}$, $4v^5_{rxn,c}$, $6v^5_{rxn,c}$ (c = 1, 2, 3, 4)
	10 bar	1.4	$v^{10}_{rxn,c}$, $2v^{10}_{rxn,c}$, $4v^{10}_{rxn,c}$, $6v^{10}_{rxn,c}$ (c = 1, 2, 3, 4)
	15 bar	0.94	$v^{15}_{rxn,c}$, $2v^{15}_{rxn,c}$, $4v^{15}_{rxn,c}$, $6v^{15}_{rxn,c}$ (c = 1, 2, 3, 4); $6v^{15}_{rxn,c}$ (c = 5, 6, 7, 8)

(*) Permeate channel inlet velocities are expressed as functions of the reaction channel inlet velocities. For example, at 5 bar, each reaction channel inlet velocity value ($v^5_{rxn,c}$) is coupled with permeate channel inlet velocities set to $v^5_{rxn,c}$, $2v^5_{rxn,c}$, $4v^5_{rxn,c}$ and $6v^5_{rxn,c}$ (c = 1, 2, 3, 4).

$$r_{RWGS} = k_0 e^{-\left(\frac{E_a}{RT}\right)} \left[y_{CO_2} y_{H_2} - \frac{y_{CO} y_{H_2O}}{K_{eq}} \right] C_{FP} \quad (3)$$

Isothermal, steady state operation of the unit cell (Fig. 1b), the repeating core geometry of the multichannel reactor shown in Fig. 1a, is modeled by the simultaneous solution of the mass and momentum conservation equations in the fluid phases of the reaction and permeate channels, reactive transport in the porous catalyst layer and inter-channel transfer of H_2O and H_2 across the SOD membrane layer. The model considers *x* and *y* coordinates as the independent variables, and the changes in the *z* coordinate, the third spatial dimension, are neglected. This simplification is based on the negligible differences between 2D and 3D models set up for a similar reaction domain and conditions [40], and offers notable reductions in the computational cost. The model equations, correlations and mixing laws needed for the estimation of the physical properties of the fluids (*e.g.*, density, viscosity, gas phase and effective diffusivities), which are assumed to follow Newtonian behavior, and the boundary conditions are presented in SI. The set of partial differential equations are solved numerically by the finite volume method (ANSYS, v. 19.2). For this purpose, zones of the unit cell (Fig. 1b) are meshed with a total of 5.76×10^5 square cells, each having an edge length of 1.25×10^{-5} m. This configuration refers to the finest meshing possible with the Gambit platform, so no additional cases are studied for the mesh independency. Reaction kinetics (Eq. (3)), binary diffusivity correlations (SI, Table S3) are integrated to the numerical solution in the form of user defined functions (UDF), each of which is called and evaluated at every nodal point.

Transport of H_2O and H_2 across the SOD membrane is modeled by creating a wall with zero thickness between the channels. In order to compute membrane mass flux, mesh cells neighboring the zero–thickness wall are defined as sink and source zones, which are assigned with source terms involving mass inflow and outflow of H_2O and H_2, respectively. Evaluation of these terms, based on Fick's Law given in Eq. (4) are coded and formulated in a UDF whose details are described elsewhere [16,17]:

$$J_i = Perm_i \left(p^i_{rxn} - p^i_{perm} \right) \quad (4)$$

The membrane UDF is integrated to the reactor model equations presented in SI (Table S2). Data needed for executing the membrane model is adapted from the literature [34]. At 473 K, the SOD membrane is reported to have high steam perm–selectivity to CO and CO_2, so permeation of carbon oxides is not considered. Cross–membrane transport of H_2 is justified by the relatively lower H_2O/H_2 selectivity of 4.6. Steam permeability of the SOD membrane is reported as 6.5×10^{-8} mol m^{-2} s^{-1} Pa^{-1} [34]. Considering that this experimental data is reported at 473 K for the membrane separation of the binary mixtures, permeability of H_2O is taken as 3×10^{-8} mol m^{-2} s^{-1} Pa^{-1} to account for (a) the multicomponent nature of the gas mixture that can cause competitive adsorption to the membrane surface and hinder permeability [41,42], and (b) the lack of experimental data at 523 K. It was shown that the contributions of promoted permeation of H_2O and H_2 on the performance of an SOD membrane–integrated microreactor become negligible when the permeability exceeds 3×10^{-8} mol m^{-2} s^{-1} Pa^{-1} [16]. This finding is used as a justification for fixing H_2O permeability to 3×10^{-8} mol m^{-2} s^{-1} Pa^{-1} in the current study. In order to quantify the benefits of integrated separation, a set of benchmark simulations are run in the absence of membrane. For this purpose, an impermeable wall is imposed simply by nullifying the source terms that account for interchannel mass transfer in the membrane model. Outcomes of the membrane–integrated simulations are also benchmarked against equilibrium CO_2 conversion computed as described in SI (Section A).

Impacts of the proposed parametric study are measured in terms of CO_2 conversion, the amount (moles of) CO_2 converted and the molar syngas composition [43] defined in Eqs. (5–7), respectively:

$$CO_2 \text{ conversion } (\%) = \frac{F_{CO_2,in} - F_{CO_2,out}}{F_{CO_2,in}} \times 100 \quad (5)$$

$$CO_2 \text{ converted } (\text{mol } s^{-1}) = F_{CO_2,in} \times CO_2 \text{ conversion (fractional)} \quad (6)$$

$$\text{Syngas } composition = \frac{F_{H_2,out} - F_{CO_2,out}}{F_{CO,out} + F_{CO_2,out}} \quad (7)$$

3. Results and discussion

3.1. Comparison of the reactor model simulations with literature–based experimental data

Prior to its use in understanding the characteristics of the membrane–integrated RWGS reaction, outcomes of the reactor model are compared with the literature–based experimental data reported for the $CuO/ZnO/Al_2O_3$ catalyst [37]. Integrated separation function is disabled in the model as the experimental study was run in a membraneless packed–bed reactor. In this regard, the reactor model quantifies the operation of the catalyst washcoated microchannel (*i.e.*, the reaction channel depicted in Fig. 1b), and comprises conservation equations formulated in two spatial dimensions at steady state, correlations and mixing rules for the estimation of fluid properties (SI, Section B) and the kinetic model (Eq. (3)). The membraneless reactor model is run at 523 K, 1 bar, H_2:CO_2 = 6 and within the $W/F_{CO_2,in}$ range of ~2–44 g_{cat} h mol^{-1} due to the availability of the experimental CO_2 conversions at these conditions. Absence of the transport resistances in both reactors allows direct comparison of the data produced on a particulate bed of catalyst with the data currently computed on the washcoated form of the same catalyst. The results shows that experimental and simulated CO_2 conversions are satisfactorily close to each other at $W/F_{CO_2,in}$ > ~28 g_{cat} h mol^{-1}; at $W/F_{CO_2,in}$= 40 g_{cat} h mol^{-1}, the residence time used in the current study, reactor model successfully captures the experimental CO_2 conversion (Fig. 2). On the other hand, the reactor model underpredicts the experimental data at $W/F_{CO_2,in}$ < ~28 g_{cat} h mol^{-1}. This deviation observed at p = 1 bar is believed to be due to the pressure correction factor (C_{FP}, defined in SI (Eq. (S2)) in the reaction rate expression (Eq. (3)). At the default pressure of 15 bar (Table 1), the mathematical model is expected to give higher CO_2 conversion closer to the experimental data as the reaction rate increases due to the change in C_{FP} from 1.0 (at 1 bar) to 3.3 (at 15 bar)). The pronounced impact of C_{FP} at 15 bar is expected to be observed at low residence times where CO_2 conversion is dictated mostly by the reaction kinetics. At residence times > ~28 g_{cat} h mol^{-1}, however, the slightly reduced rate of increase in CO_2 conversion indicates the impact of thermodynamic equilibrium which limits the contribution of C_{FP}. Therefore, the overlap between the experimental and predicted CO_2 conversions at $W/F_{CO_2,in}$= 40 g_{cat} h mol^{-1} shown for 1 bar (Fig. 2) is expected to remain almost unaffected at 15 bar. These finding prove the capability of the reactor model in successful capture of its physical

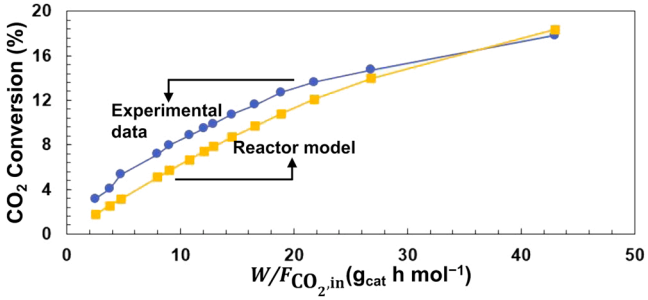


Fig. 2. Comparison of the membraneless microchannel reactor model results with the experimental data adopted from [37] (T = 523 K, p = 1 bar, H_2:CO_2 = 6).

counterpart and allows its use for the parametric analysis of the membrane–integrated RWGS reaction.

3.2. Effects of reactor pressure, reactor and permeate channel inlet velocities

The effects of the velocities of the reactive mixture ($H_2 + CO_2$) and sweep (H_2) streams on the reactor performance are investigated at pressures of 5, 10 and 15 bar. Combinations of the reactive mixture and sweep gas inlet velocities are presented in Table 1. For example, at 5 bar, each reaction channel inlet velocity value ($v^5_{rxn,c}$) is coupled with permeate channel inlet velocities set to $v^5_{rxn,c}$, $2v^5_{rxn,c}$, $4v^5_{rxn,c}$ and $6v^5_{rxn,c}$. For benchmarking purposes, computations are also done for the membraneless case (labeled as "NM" in Fig. 3). The findings clearly show the contribution of membrane assisted efflux/influx of H_2O/H_2 from/to the reaction channel on CO_2 conversion, which remains almost unchanged with the inlet velocity of the reactive mixture in the absence of membrane (Fig. 3). In other words, elongating the contact of $H_2 + CO_2$ mixture does not improve CO_2 conversion of ~16% which is close to the thermodynamic limit of 16.8% (SI, Section A). Contribution of the membrane manifests itself in > 2 times increase in conversion, with the extent of increase being more notable at higher $W/F_{CO_2,in}$ values (*i.e.*, inlet velocities decreasing from $v^5_{rxn,4}$ to $v^5_{rxn,1}$ (Fig. 3a) and from $v^{10}_{rxn,4}$ to $v^{10}_{rxn,1}$ (Fig. 3b)). The results also reveal that further increase in the sweep gas inlet velocity keeps on promoting CO_2 conversion. These findings can be explained by two specific phenomena induced by transmembrane mass transfer. Owing to its higher partial pressure in the permeate channel, H_2 diffuses into the reaction channel to promote RWGS reaction kinetically (as quantified by Eq. (3)) and thermodynamically. The resulting increase in CO_2 conversion favors production of H_2O and elevates its partial pressure in the reaction zone. Since the sweep gas is pure H_2, the positive partial pressure gradient allows transmembrane efflux of H_2O from the reaction channel, leading to further promotion of the kinetic and thermodynamic effects to boost CO_2 conversion. These phenomena become more pronounced at higher amounts of sweep gas (H_2) as the permeate side H_2O becomes less concentrated, which maintains the driving force for H_2O transport across the membrane layer. Fig. 3a, b also points out the decline in membrane–assisted CO_2 conversion at reduced residence times (*i.e.*, higher inlet velocities) of the reactive mixture. For example, at the highest permeate channel inlet velocities (denoted as $6v^5_{rxn,c}$ and $6v^{10}_{rxn,c}$ for 5 and 10 bar, respectively), conversion decreases from ~43% to ~25% (at 5 bar) and from ~48% to ~30% (at 10 bar). The decrease can be explained by the reduced contact of species with the membrane and catalyst layers, which eventually limits the rate of reaction and cross membrane transport. Comparison of Fig. 3a and b provides insight into the role of reactor pressure and shows slight improvement in conversion upon 5 bar increase. As explained in Section 2, both channels operate at the same pressure. It is worth noting that, for a given residence time, the molar flow rate of $H_2 + CO_2$ mixture to the reaction channel is kept the same by balancing the pressure change with inlet volumetric (and linear) flow rate. In this regard, the slightly higher conversions in favor of 10 bar can be related to the widened partial pressure difference for interchannel H_2 transport due to higher H_2 partial pressure on the permeate side.

According to Fig. 3a, b, CO_2 conversion can be maximized to ~43% and ~48% at 5 and 10 bar, respectively, when the membrane reactor is operated at the lowest residence time of the reactive mixture and the highest permeate channel inlet velocity. While the simulations point out the possibility of obtaining CO_2 conversions ~3 times higher than the thermodynamic limit, the advantage is masked by the lowest amount of CO_2 converted (Fig. 3c, d). Obviously, this scenario is unwanted from a capacity point of view; the reactor block volume increases due to the higher number of reaction and permeate channels for delivering a targeted syngas throughput or, in the case of constant reactor volume (*i.e.*, fixed number of channels), syngas production decreases. In this context, reactor operation that couples the highest flow rate of $H_2 + CO_2$ mixture with the highest flow rate of the H_2 sweep gas maximizes the amount of CO_2 converted and can be regarded as the case requiring the lowest

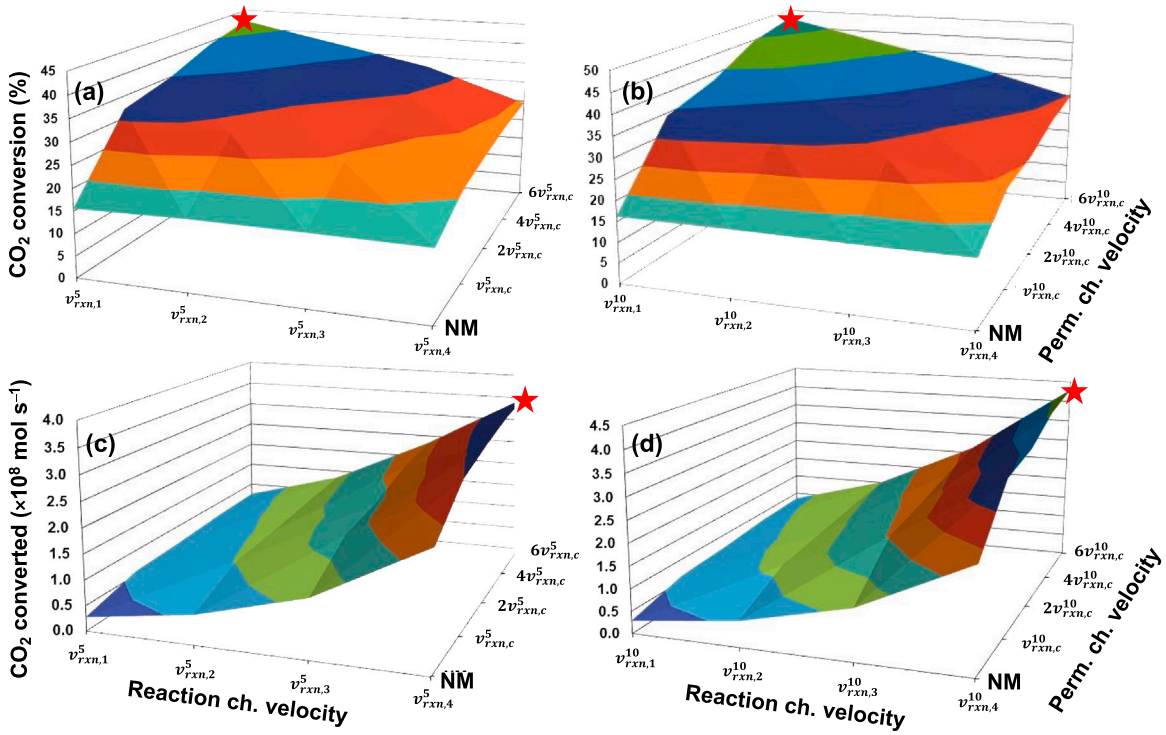


Fig. 3. CO_2 conversion (a, b) and amount of CO_2 converted (c, d) as functions of reaction and permeate channel inlet velocities. Results in (a) and (c) are computed at 5 bar; (b) and (d) are computed at 10 bar. "NM" refers to the membraneless case. Refer to Table 1 for the values of reaction and permeate channel inlet velocities (T = 523 K, H_2:CO_2 = 3, flow direction in membrane–integrated cases: co–current).

reactor volume for a given capacity objective.

For further elaboration of the effects of inlet velocities and reactor pressure, simulations are extended to 15 bar where broader range of reactant inlet velocities are studied. In this regard, residence time range of 5–20 g_{cat} h mol^{-1} is investigated in addition to 40–320 g_{cat} h mol^{-1} studied in 5 and 10 bar cases. The range extension is studied only at 15 bar, the highest pressure in the current study, due to the positive effect of pressure on membrane separation (Fig. 3). In the 5–20 g_{cat} h mol^{-1} range, permeate channel inlet velocity is set to 6 times the pertinent reaction channel inlet velocity and the results are benchmarked with the membraneless case. Coupling of the reactant/sweep inlet velocity corresponding to the residence time range of 40–320 g_{cat} h mol^{-1} is carried out as described for the 5 and 10 bar simulations (Fig. 3). Fig. 4a reveals that, membrane aided CO_2 conversion at 15 bar can be maximized to ~52%, which exceeds ~48% and ~43% obtained at 10 and 5 bar, respectively. However, decreasing residence time (*i.e.*, increasing $H_2 + CO_2$ mixture inlet velocity from $v^{15}_{rxn,1}$ to $v^{15}_{rxn,4}$) reduces the contribution of membrane separation to CO_2 conversion as evident from its reduction from ~52% to ~32% (Fig. 4a). This trend continues at residence times < 40 g_{cat} h mol^{-1} until the case ($v^{15}_{rxn,8}$ corresponding to 5 g_{cat} h mol^{-1}) where the impact of SOD membrane separation diminishes to end up with CO_2 conversion < 10% (Fig. 4b). On the other hand, this extreme case delivers the highest molar flow rate of CO_2 converted due to its positive correlation with the reaction channel inlet velocity (Fig. 4c, d).

In addition to per cent CO_2 conversion and molar amount of CO_2 transformed, syngas composition, defined in Eq. (7), is used as another metric to evaluate membrane–integrated reactor performance. The ratio in Eq. (7) is used for quantifying syngas input for high temperature FT synthesis, methanol and DME production, for which the desired syngas composition is ~2 [43]. Therefore, Eq. (7) is evaluated for all combinations of the reaction channel inlet velocity and reactor pressure. Considering the results in Figs. 3 and 4, only the highest permeate channel inlet velocity (= 6 × the pertinent inlet velocity of the $H_2 + CO_2$ mixture) is considered. The results presented in Fig. 5 point out decrease in the value of Eq. (7) with increase in reactive mixture inlet velocity (*i.e.*, decreasing residence time) and increase in the value of Eq. (7) with reactor pressure, with the extent of increase being more pronounced at higher residence times. The highest ratio of 9.8 is a result of the maximized transmembrane transport of H_2 into and H_2O from the reaction channel. When these are combined with the maximum CO_2 conversion (Figs. 3a, b, 4a), numerator of Eq. (7) is maximized, whereas the denominator remains almost unchanged due to the balance between CO throughput and unconverted CO_2. In the other extreme simulated at $v^{15}_{rxn,8} = 7.5 \times 10^{-2}$ m s^{-1} (equivalent to 5 g_{cat} h mol^{-1}) and 15 bar, syngas composition is minimized to 2.3 (Fig. 5). This value is almost equal to that of the membraneless case owing to the negligible effect of the membrane discussed in Fig. 4b, d. The change between these extremes, however, is not linear; at 15 bar the decrease in composition is more notable in the $v^{15}_{rxn,1-4}$ range (also common to 5 and 10 bar cases, with the successive changes being larger at higher pressures), but becomes only incremental at $v^{15}_{rxn,4-8}$ (Fig. 5). This pattern provides useful insight into identification of the operating conditions offering plausible balance between the metrics defined in Eqs. (5) and (6). In this regard, setting the reaction channel inlet velocity to $v^{p}_{rxn,4}$ (the default value corresponding to $W/F_{CO_2,in}$ = 40 g_{cat} h mol^{-1} (Table 1)) and sweep gas inlet velocity to 6 × $v^{p}_{rxn,4}$ can offer syngas composition of 2.7–3.8. At 5 bar, CO_2 conversion is 24.5% which exceeds ~16% of the membraneless case (Fig. 3a), and the amount of CO_2 converted is 3.5 × 10^{-8} mol s^{-1} which is ~53% above of the membraneless case (Fig. 3c). These conditions can be regarded as optimum. CO_2 conversion is significantly hampered when dosing the reactive mixture above $v^{p}_{rxn,4}$ (Fig. 4b). In this scenario, the capital and operating expenses associated with the recycling the unconverted CO_2 and H_2 increase due to the reduced effectiveness of the *in–situ* membrane separation. On the other hand, the overall reactor block volume, catalyst mass and membrane surface area needed to process a certain $H_2 + CO_2$ mixture capacity increase at < $v^{p}_{rxn,4}$ due to

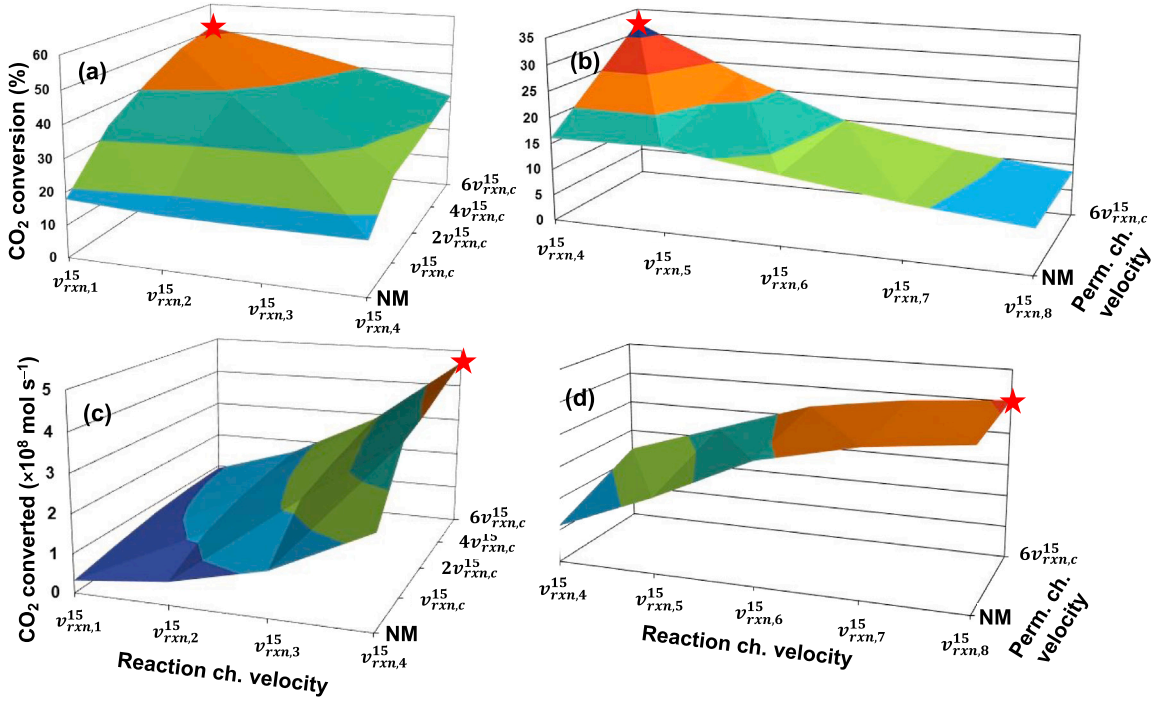


Fig. 4. CO_2 conversion (a, b) and amount of CO_2 converted (c, d) as functions of reaction and permeate channel inlet velocities. In (a) and (c) permeate channel inlet velocities equal to 1, 2, 4 and 6 times the pertinent reaction channel inlet velocity are studied. In (c) and (d) permeate channel inlet velocity equal to 6 times the pertinent reaction channel inlet velocity is studied. "NM" refers to the membraneless case. Refer to Table 1 for the values of reaction and permeate channel inlet velocities (T = 523 K, p = 15 bar, H_2:CO_2 = 3, flow direction in membrane–integrated cases: co–current).

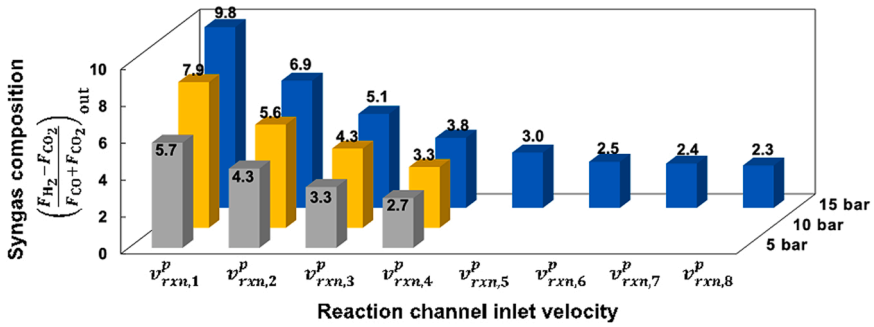


Fig. 5. Syngas composition (defined in Eq. (7)) as functions reaction channel inlet velocity ($v^p_{rxn,c}$) and reactor pressure (p = 5, 10, 15 bar). In all cases, permeate channel velocity is set to 6 × $v^p_{rxn,c}$ (T = 523 K, H_2:CO_2 = 3, flow direction: co–current. Refer to Table 1 for the values of the reaction channel inlet velocities).

the decreasing amount of CO_2 converted (Fig. 3c, d; 4c, d) as the moles of reactive mixture converted per reaction channel gets lesser at higher values of $W/F_{CO_2,in}$. In this case, more reaction and permeate channels are needed to match the desired operational capacity, which elevates capital expenditure related to the material (reactor material, catalyst and membrane) and labor (micromachining and bonding of the plates, catalyst coating, decoration of the membrane layers) needed for manufacturing and installation of the multichannel reactor block. Additionally, the operation becomes more complex due to the requirement of equipartitioning of the feed to the reaction and permeate channels. The economic analysis presented herein is limited to a brief qualitative discussion, and further quantification is out of the scope of the current work.

3.3. Effects of molar inlet H_2:CO_2 ratio and flow partitioning

The effect of the molar composition of the reactive mixture (H_2:CO_2) on CO_2 conversion and syngas composition are presented in Fig. 6. The ratio is tuned by proper modification of the molar inlet H_2 and CO_2 flow rates to keep the total molar inlet flow rate constant. For comparison purposes, simulations of the membrane–integrated cases at H_2:CO_2 = 3 are benchmarked with the membraneless counterparts. In accordance with the findings in Section 3.2, membrane contribution to CO_2 conversion and the magnification of the contribution with pressure are evident in Fig. 6. Nearly identical conversions computed at 5 and 15 bar in the absence of the membrane are due to the clear effect of the thermodynamic barrier (= 16.8%).

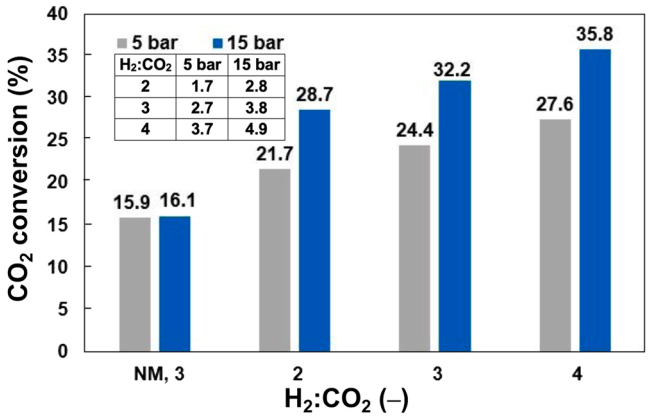


Fig. 6. Effect of molar inlet H_2:CO_2 ratio on CO_2 conversion at 5 and 15 bar. In membrane–integrated cases, permeate channel inlet velocity = 6 × pertinent reaction channel inlet velocity. "NM" refers to the membraneless case (T = 523 K, H_2:CO_2 = 3, flow direction in membrane–integrated cases: co–current. Reaction channel inlet velocity = 2.8 ×10^{-2} and 9.4 ×10^{-3} m s^{-1} at pressures of 5 and 15 bar, respectively). Syngas composition (Eq. (7)) as a function of pressure and H_2:CO_2 ratio is shown in the inset table.

Regardless of the reactor pressure, H_2 enrichment of the reactive mixture manifests itself with monotonic increase in CO_2 conversion. When combined with the transmembrane H_2 supply, elevated H_2:CO_2 ratios lead to substantial increase in H_2 partial pressure that promotes the rate of RWGS (Eq. (3)) and drives the reaction thermodynamically in the forward direction. As discussed in Section 3.2, higher CO_2 conversions in 15 bar operation come from the increased extent of H_2 supply from the permeate side. The effect of H_2:CO_2 ratio on syngas composition is presented in the inset table in Fig. 6. In contrast with its positive effect on conversion, elevating H_2:CO_2 causes syngas composition to increase undesirably when the ideal value of ~2 is considered. Along these lines, operation at 5 bar and H_2:CO_2 = 2 is worth to consider as this configuration is capable of delivering syngas composition equal to 1.7 (Fig. 6), while offering 21.7% conversion which is ~1.6 times higher than its counterpart (13.7%, not shown in Fig. 6) computed in the membraneless reactor run at H_2:CO_2 = 2 under identical conditions. Moreover, the composition of 2.8 obtained at H_2:CO_2 = 2 and 15 bar should not be overlooked considering the pressurization requirements of syngas in its post–utilization. These findings point out the possibility of customizing syngas composition by the RWGS reaction run in the proposed membrane–integrated reactor configuration.

In addition to the molar composition of the H_2 + CO_2 mixture, the effect of flow directions of the reactive mixture and sweep gas on reactor performance is investigated. In this regard, co–current flow pattern, the default mode of flow direction in this study, is compared with the counter–current flow operation at the minimum and maximum values of the reactor pressure and feed composition (5, 15 bar; H_2:CO_2 = 2, 4) to elucidate the impact of flow partitioning in a broader range of operating conditions. The results, presented in Fig. 7, show that reactor operation at 5 bar remains insensitive to the flow direction regardless of the H_2:CO_2. At 15 bar, however, dosing the streams at opposite directions seems to give a slight improvement in CO_2 conversion, with the extent of increase being marginally higher at H_2:CO_2 = 4. The pattern in favor of the counter–current flow at 15 bar is most likely related with enriching

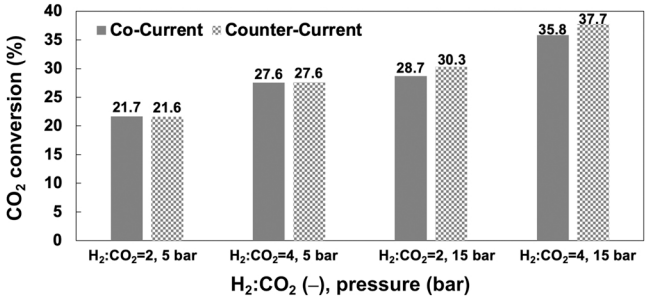


Fig. 7. Effect of flow direction on CO_2 conversion. Permeate channel inlet velocity = 6 × pertinent reaction channel inlet velocity. (T = 523 K. Reaction channel inlet velocity = 2.8 ×10^{-2} and 9.4 ×10^{-3} m s^{-1} at pressures of 5 and 15 bar, respectively).

the reactive mixture with transmembrane H_2 influx at the downstream of the reaction channel where H_2 consumption due to reaction is more significant and the local driving force for H_2 transport (*i.e.*, difference between the H_2 partial pressures at the permeate and reaction channels) is inherently high. Similarly, the local H_2O partial pressure gradient is higher at the downstream as there is no steam at the inlet of the permeate channel. Nevertheless, these phenomena have no contribution to the reactor operation at 5 bar (Fig. 7). These findings point out that flow direction of the reactive and sweep streams has no major implications on the reactor performance under the current conditions. However, marginal improvement at 15 bar indicates counter–current flow as a better operational configuration in case of running membrane–assisted RWGS reaction at pressures in excess of 15 bar.

3.4. Comparison of the packed–bed and microchannel reactor configurations

The results discussed in the previous sections highlight the potentials of the membrane–integrated microreactor in overcoming thermodynamic limitations and running RWGS at 523 K which is below the temperatures reported for the conventional catalytic route [6–8]. In an attempt for understanding the effect of the reactor architecture on CO_2 utilization *via* RWGS, the current configuration (Fig. 1) is benchmarked with a packed–bed membrane reactor (PBMR) under equivalent conditions. Comparison is based on the residence time ($W/F_{CO_2,in}$) of 40 g_{cat} h mol^{-1} and carried out at 523 K, 5 bar and H_2:CO_2 = 3. Details regarding sizing and modeling of the microchannel and PBMR are provided in Section 2 and SI (Section D), respectively.

The results, presented in Fig. 8, show that the microchannel membrane reactor outperforms the PBMR; the former offers almost ~2.5 times higher molar CO_2 consumption per mass of catalyst and reactor volume, and delivers ~33% higher CO_2 conversion. Considering that both reactors are modeled to run without being masked by transport limitations, the differences in the performance can be attributed to the higher membrane surface area per unit volume in the microchannel unit. The resulting elongated contact of H_2O and H_2 with the membrane layer magnifies the benefits of interchannel species transfer on the kinetics and thermodynamics of RWGS reaction discussed in Section 3.1. Increased geometric accessibility of the membrane is closely correlated with the stronger (more than one order of magnitude higher) volumetric intensification offered by the sub–millimeter structuring of the flow paths [29]. Combined with the advantage of facile scalability through numbering up strategy and the possibility of customizing syngas composition by careful selection of the operating conditions (Section 3.2), the findings in Fig. 8 point out membrane–integrated microreactor driven RWGS as an efficient route for syngas production at the desired scale.

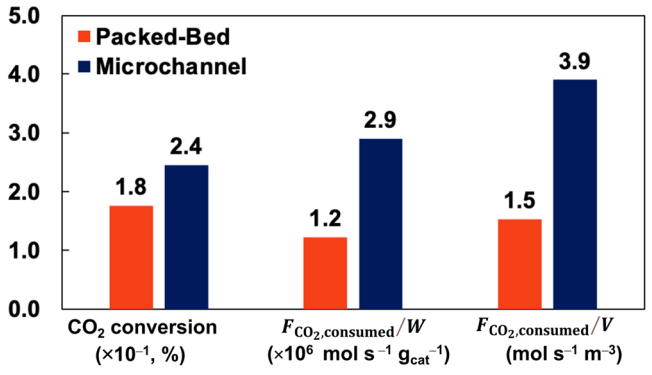


Fig. 8. Comparison of the microchannel membrane reactor with a PBMR ($W/F_{CO_2,in}$ = 40 g_{cat} h mol^{-1}, T =523 K, p =5 bar, H_2:CO_2 = 3).

4. Conclusions

Catalytic conversion of CO_2 and H_2 mixture to syngas *via* RWGS is modeled isothermally in a membrane separation integrated microchannel reactor. The reactor model is shown to capture the literature–based experimental data successfully. Due to the permselectivity of the SOD membrane to H_2O and H_2, the proposed intensified reactor is configured in a way to maximize CO_2 transformation by simultaneous transmembrane H_2 dosing into and H_2O removal from the reaction medium. The findings, all obtained at 523 K, show the effects of the operating conditions on per cent CO_2 conversion, the molar amount of CO_2 converted and syngas composition as summarized below:

- Fluid velocities at the reaction and permeate channel inlets affect reactor performance significantly. CO_2 conversion can be elevated at lower inlet velocities (*i.e.*, longer residence time) of the H_2 + CO_2 mixture and higher inlet velocities of the sweep gas (H_2). These conditions favor the extent of cross–membrane H_2 supply that promotes RWGS kinetically and thermodynamically. The promotion is magnified by the ejection of product H_2O into the permeate side.
- The maximum CO_2 conversion of 52% is obtained at 15 bar, H_2:CO_2 = 3 at which the thermodynamic limit is 16.8%. This superior performance is observed at minimized molar dosing of the H_2 + CO_2 mixture, which undesirably limits the amount of CO_2 converted and reduces the syngas production capacity of the multichannel reactor. Both performance metrics are positively correlated with reactor pressure and the inlet velocity (*i.e.*, molar flow rate) of the sweep gas.
- Membrane integration introduces significant flexibility in molar composition of syngas which changes in the 2.3–9.8 range. The composition decreases by reducing reactor pressure and increasing the inlet velocity of the reactive mixture. Imposing the recommended value of ~2 (for high temperature FT synthesis, methanol and DME production) points out the most favorable set of operating conditions that can deliver syngas composition of 2.7 ($W/F_{CO_2,in}$ = 40 g_{cat} h mol^{-1}; sweep gas inlet velocity = 6 × the pertinent reaction channel inlet velocity, 5 bar, H_2:CO_2 = 3). Under these conditions, CO_2 conversion is 24.5% which exceeds ~16% of the membraneless case, and the amount of CO_2 converted is 3.5 × 10^{-8} mol s^{-1} which is ~53% above of the membraneless case.
- Syngas composition can be decreased further by reducing molar inlet H_2:CO_2 ratio. Despite a limited loss in CO_2 conversion, decreasing H_2:CO_2 from 3 to 2 allows syngas production in a composition range of 1.7–2.8 at 5 and 15 bar, respectively. The conclusions stated so far for the co–current flow of the reactive mixture and sweep streams remain almost unchanged when the dosing is carried out in the counter–current mode. Nevertheless, the negligible impact of flow partitioning is valid for the current set of operating conditions and is likely to change in favor of the counter–current flow at higher reactor pressures.
- Under equivalent operating conditions, the micro structured unit proposed herein can deliver ~2.5 times higher molar CO_2 consumption per mass of catalyst and reactor volume and ~33% higher CO_2 conversion than a packed–bed membrane reactor.

CRediT authorship contribution statement

Mert Can Ince: Data curation, Investigation, Methodology, Software, Visualization, Writing – original draft. **H. Hasan Koybasi:** Data curation, Conceptualization, Investigation, Methodology, Software. **Ahmet K. Avci:** Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgement

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.cattod.2023.114130.

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